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Enhancing predictive accuracy for urinary tract infections post-pediatric pyeloplasty with explainable AI: an ensemble TabNet approach

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Ureteropelvic junction obstruction (UPJO) is a common pediatric condition often treated with pyeloplasty. Despite the surgical intervention, postoperative urinary tract infections (UTIs) occur in over 30% of cases within six months, adversely affecting recovery and increasing both clinical and economic burdens. Current prediction methods for postoperative UTIs rely on empirical judgment and limited clinical parameters, underscoring the need for a robust, multifactorial predictive model. We retrospectively analyzed data from 764 pediatric patients who underwent unilateral pyeloplasty at the Children's Hospital affiliated with the Capital Institute of Pediatrics between January 2012 and January 2023. A total of 25 clinical features were extracted, including patient demographics, medical history, surgical details, and various postoperative indicators. Feature engineering was initially performed, followed by a comparative analysis of five machine learning algorithms (Logistic Regression, SVM, Random Forest, XGBoost, and LightGBM) and the deep learning TabNet model. This comparison highlighted the respective strengths and limitations of traditional machine learning versus deep learning approaches. Building on these findings, we developed an ensemble learning model, meta-learner, that effectively integrates both methodologies, and utilized SHAP(Shapley Additive Explanation, SHAP) to complete the visualization of the integrated black-box model. Among the 764 pediatric pyeloplasty cases analyzed, 265 (34.7%) developed postoperative UTIs, predominantly within the first three months. Early UTIs significantly increased the likelihood of re-obstruction (*P* < 0.01), underscoring the critical impact of infection on surgical outcomes. In evaluating the performance of six algorithms, TabNet outperformed traditional models, with the order from lowest to highest as follows: Logistic Regression, SVM, Random Forest, XGBoost, LightGBM, and TabNet. Feature engineering markedly improved the predictive accuracy of traditional models, as evidenced by the enhanced performance of LightGBM (Accuracy: 0.71, AUC: 0.78 post-engineering). The proposed ensemble approach, combining LightGBM and TabNet with a Logistic Regression meta-learner, achieved superior predictive accuracy (Accuracy: 0.80, AUC: 0.80) while reducing dependence on feature engineering. SHAP analysis further revealed eGFR and ALB as significant predictors of UTIs post-pyeloplasty, providing new clinical insights into risk factors. In summary, we have introduced the first ensemble prediction model, incorporating both machine learning and deep learning (meta-learner), to predict urinary tract infections following pediatric pyeloplasty. This ensemble approach mitigates the dependency of machine learning models on feature engineering while addressing the issue of overfitting in deep learning-based models like TabNet, particularly in the context of small medical datasets. By improving prediction accuracy, this model supports proactive interventions, reduces postoperative infections and re-obstruction rates, enhances pyeloplasty outcomes, and alleviates health and economic burdens.

Level of evidence IV **Case series with no comparison group.**

Keywords Urinary tract infection, Unilateral pyeloplasty, Machine learning, Deep learning, TabNet, Ensemble model, Interpretable prediction model

Abbreviations

UPJO	Ureteropelvic junction obstruction
VUR	Vesicoureteral reflux syndrome
UTI	Urinary tract infection
APD	Anterior posterior diameter
OP	Open pyeloplasty
LP	Laparoscopic pyeloplasty
BMI	Body max index
BUN	Blood urea nitrogen
eGFR	Estimated glomerular filtration rate
ALB	Albumin
GLB	Globulin
UA	Urine acid
Cr	Serum creatinine, SCR
BUN	Blood urea nitrogen
N%	Neutrophilicgranulocyte, NEUT%
L%	Lymphocyte%
ROC	Receiver operating characteristic
AUC	Area under the curve
SHAP	Shapley Additive Explanation

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Ureteropelvic junction obstruction (UPJO) is defined as an obstruction at the junction of the kidney and ureter, resulting in decreased urine flow from the pelvis to the ureter. UPJO is one of the main causes of infantile hydronephrosis. UPJO (Ureteropelvic Junction Obstruction) is the most common cause of pathological hydronephrosis in children, with an incidence rate of approximately 1:1500[1,2]. Approximately 20–50% of children with UPJO ultimately require surgical intervention[3]. If left untreated, it may cause hydronephrosis, chronic infection or urolithiasis, and can often lead to progressive renal insufficiency[4].

However, despite its high success rate, surgical treatment of UPJO is not free from complications. The main complications after surgical treatment of UPJO are urinary tract infections (UTI), urinary extravasation and leakage, pyelonephritis, bleeding and recurrent UPJO[5]. Multiple studies have demonstrated that UTI increases the risk of anastomotic restenosis following pyeloplasty in children with UPJO, and is considered one of the significant factors contributing to postoperative anastomotic restenosis[6–9]. Incidence of UTI remains high and challenging for pediatric urologists. UTI, which is mostly caused by Escherichia coli, is one of the most common types of infection in children[10], which is more common in boys (3.7%) than in girls (2%) in the first year of life, and more common in girls after infancy[11]. Previously, some studies have described risk factors for UTI in children, including male sex, weight, BUN level, recurrence of UTI within 3 months, long-term catheter retention, double-J catheter retention, bilateral double-J catheter retention, etc[12,13]. However, the development of predictive models for UTI following unilateral pyeloplasty is very limited, especially in large-sample cohort studies. Similarly, there are currently no artificial intelligence models for UTI after pyeloplasty.

The healthcare industry has amassed a substantial amount of data, including medical records and patient diagnostic outcomes.In light of the recent advancements in artificial intelligence, encompassing machine learning (ML) and deep learning (DL), these technologies are now being increasingly employed in clinical decision-making processes[14]. With the advent of high-efficiency ensemble learning models[15,16], machine learning (ML) has achieved remarkable breakthroughs in accurately and efficiently processing tabular data, with applications in clinical medicine being particularly amenable to these methods. The application of machine learning (ML) algorithms has demonstrated promising predictive performance in the fields of kidney diseases and urothelial tumors[17–19]. However, there are still few reports on its use in functional urology[20]. A comprehensive review published in The Lancet in 2024 demonstrated that healthcare professionals and patients perceive machine learning (ML) risk prediction models as capable of enhancing benefits within the healthcare domain[21].

In recent years, with the continuous advancement of deep learning technologies, deep learning network models represented by CNNs and Transformers have gradually been employed in the diagnosis and prognosis of diseases. TabNet is an end-to-end deep learning-based architecture equipped with an encoder. It establishes a sequential multi-step architecture, which aids in the decision-making process by selecting relevant features. In recent years, TabNet has increasingly been employed in clinical prediction applications[22–26]. However, the application of deep neural networks (DNNs), exemplified by TabNet, in medicine is still significantly less prevalent compared to traditional machine learning models. This disparity may be attributed to the predominance of small, single-center datasets in medical data structures.

In real-world clinical scenarios, predicting UTIs following pediatric pyeloplasty is particularly challenging due to the rarity of the condition and the high exclusion rate of subjects, which together result in a small sample size. This limitation underscores the inadequacy of relying solely on either traditional machine learning (ML) or deep learning (DL) methods. We observed that while ML models are generally robust with small datasets, they require extensive manual feature engineering. Conversely, deep neural networks (DNNs), such as TabNet, can automate feature learning but are prone to overfitting and bias when applied to small datasets.

To address these challenges, we have developed a novel ensemble learning method, named Meta-Learner, which strategically combines the complementary strengths of both ML and DL. Our Meta-Learner dynamically adjusts the contributions of different base models, enabling a collaborative synergy between ML and DL approaches. This dynamic knowledge integration significantly enhances the prediction accuracy for UTIs post-pyeloplasty, offering a more reliable and robust solution compared to single-method models. This work introduces the first clinical prediction model specifically designed for UTI risk following pyeloplasty, marking a significant advancement in early diagnosis and timely intervention. By predicting UTIs more accurately, our model has the potential to reduce the long-term functional impacts of these infections, thereby improving patient outcomes. Furthermore, the integration of SHAP for interpretability ensures that the model's predictions are transparent and clinically actionable, offering insights into the relative importance of different features. This transparency is critical for gaining clinical trust and facilitating the implementation of the model in practice.

Materials and methods

Study population and data collection

This retrospective study was reviewed and approved by the Institutional Review Board of the Capital Institute of Pediatrics. Due to the retrospective nature of the study and all administrative procedures being part of routine treatment and care, the requirement for patient informed consent was waived. The study was reviewed by the institutional ethics committee, with the ethical approval number: SHERLLM2023041. Data were collected from June 1, 2023, to December 31, 2023, yielding 1033 cases of pediatric hydronephrosis. After applying the inclusion and exclusion criteria, a total of 764 cases were included in the cohort, and the study workflow is shown in Fig. 1. Patients were divided into two cohorts at an 8:2 ratio: one training set ($n = 612$) and one validation set ($n = 152$).

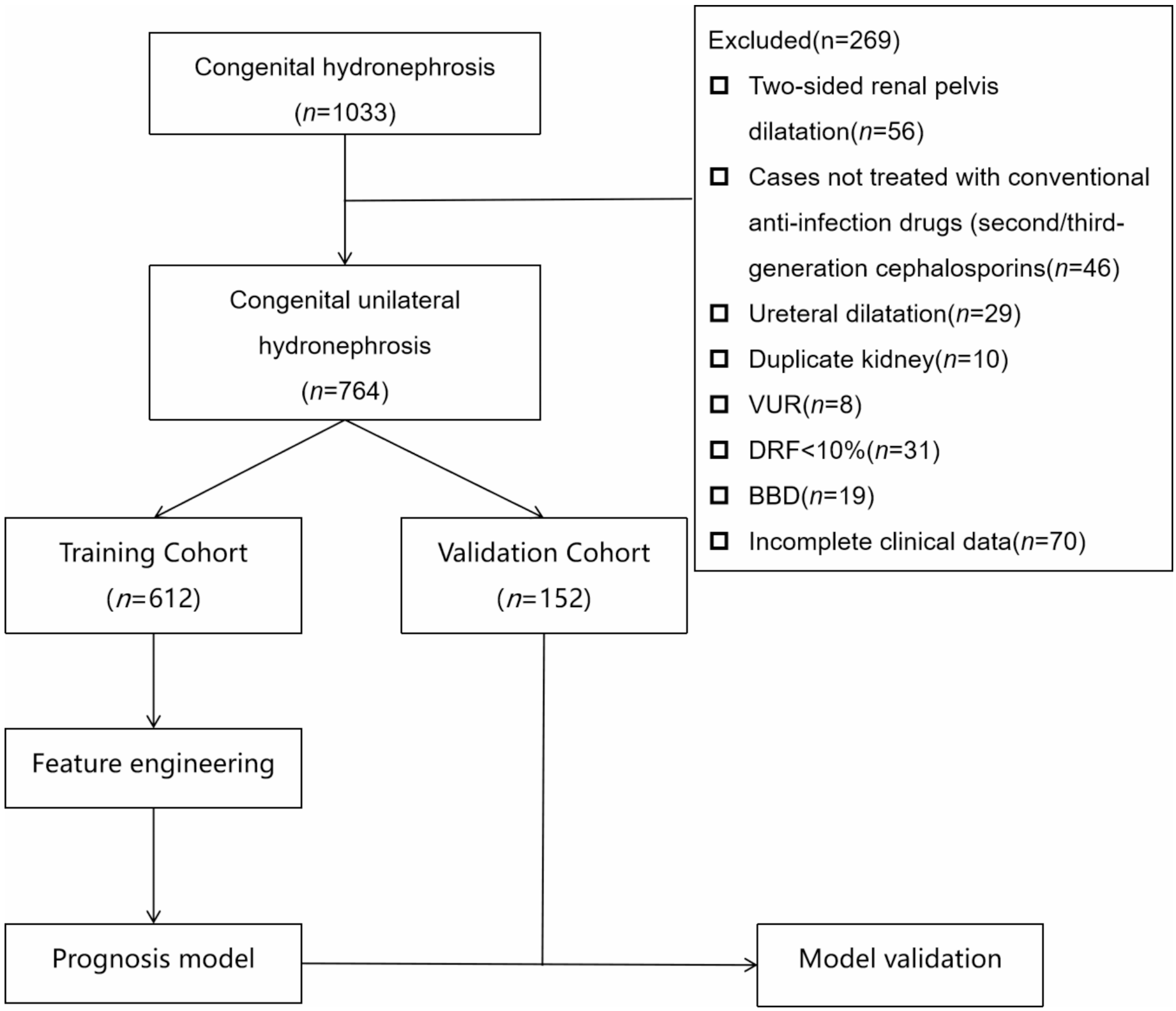


Fig. 1. Flow chart for patient selection.

The collected data for the enrolled cases included general patient information (gender, age, height, weight, Body Mass Index (BMI), preoperative conditions before pyeloplasty (abdominal pain, nephrostomy, urinary tract infections), surgical details (number of surgeries (primary/revision), operation time, surgical approach (laparoscopic/open), side of hydronephrosis), anteroposterior diameter of the renal pelvis and renal parenchymal thickness measured by preoperative Doppler color ultrasound, causes of ureteropelvic junction obstruction (congenital stenosis, aberrant vessel compression, ureteral polyps, ureteral valves, high insertion of ureter, ureteral disruption), postoperative drainage method (nephrostomy/D-J stent), immunological indicators (total protein, albumin, globulin), renal function indicators (serum creatinine, blood urea nitrogen, glomerular filtration rate, serum uric acid), and hematological indices (neutrophil ratio, lymphocyte ratio).

The outcome label of this study is clinical urinary tract infection (UTI) diagnosed after pyeloplasty, with the occurrence time defined as UTI occurring within six months after surgery.Diagnosis of urinary tract infections is based on:①The 2015 European Association of Urology (EAU) Guidelines on Urinary Tract Infections[27]. ②The 2016 Chinese Pediatric Society Nephrology Group. Evidence-based guidelines for the diagnosis and treatment of urinary tract infections[28].

Prophylactic anti-infective measures prior to surgery involved the administration of a single intravenous dose of Cefuroxime Sodium (20 mg/kg) within 30 min before the procedure. Postoperatively, Cefuroxime Sodium was administered intravenously for a duration of 5 to 7 days, with a daily dosage ranging from 30 to 100 mg/kg, divided into 3 to 4 separate doses. Adjustments to the subsequent treatment regimen were made based on clinical symptoms and routine urine monitoring. During the indwelling period of the D-J stent, in the absence of specific guidelines or consensus for prophylactic anti-infection in pediatric D-J stent placements, our center followed the recommendations outlined in the EAU and AUA guidelines for VUR and the expert consensus of the Chinese Society of Pediatric Urology[29–31]:①For infants under 2 months of age: Cefaclor suspension, administered orally once daily before bedtime, at a dosage equaling one-third of the therapeutic dose.②For children aged between 2 months and 7 years: Co-trimoxazole (Compound Sulfamethoxazole), administered at a dosage equaling one-third of the therapeutic dose.③For individuals older than 7 years: Amoxicillin and Clavulanate Potassium, administered orally once daily before bedtime, at a dosage equaling one-third of the therapeutic dose. Throughout the D-J stent placement period, routine urine tests and urinary cytology smears were conducted every two weeks to facilitate timely adjustments to the anti-infective regimen and to monitor for antimicrobial resistance.

Inclusion and exclusion criteria

Inclusion criteria: (1) Patients with unilateral hydronephrosis undergoing pyeloplasty; (2) Diagnosis of ureteropelvic junction obstruction in accordance with the European Association of Urology (EAU) guidelines and the 'Consensus on the Diagnosis and Treatment of Congenital Hydronephrosis'; (3) Complete clinical data and cooperation with postoperative follow-up.

Exclusion criteria: (1) Bilateral hydronephrosis; (2) Not treated with conventional anti-infection regimens; (3) Full-length ureteral dilation; (4) Duplicated kidney; (5) Concomitant vesicoureteral reflux; (6) Preoperative differential renal function (DRF) < 10%; (7) Children with bladder and bowel dysfunction (BBD); (8) Incomplete clinical data.

Developement of predictive model building

This study employed six machine learning and deep learning algorithms to develop predictive models for urinary tract infections following pyeloplasty. The performance of different algorithmic models was compared, and a novel ensemble learning approach was introduced (Sect. Meta learner), which combines the advantages of traditional machine learning and deep learning methods (Sect. TabNet) to enhance the performance of the predictive models. Finally, SHAP was used to visualize the integrated black-box models (Sect. SHAP analysis). The study utilized five common machine learning algorithms: Logistic Regression, Support Vector Machine (SVM), Random Forest, Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM). Model prediction accuracy and conformity were evaluated using Accuracy, Precision, Recall, and the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC). Five-fold cross-validation was used to assess discrimination and calibration. Figure 2 shows the modeling and optimization process of this study. Figure 3 shows the clinical application of this model.

accuracy= $\frac{TP+TN}{P+N}$

precision= $\frac{TP}{TP+FP}$

recall= $\frac{TP}{TP+FN}$

TP, TN, FP, and FN represent true positive, true negative, false pos.
itive, and false negative, respectively.

TabNet

TabNet is a deep neural network tailored for tabular data processing. Its encoder architecture operates in multiple sequential steps manner. At each step i, processed information from the previous step i-1 guides feature selection and generates an aggregated feature representation for overall decision-making. The model inputs batches of data with a specified batch size B and D-dimensional features without global feature selection or normalization.

TabNet represents an advancement over traditional methods such as random forests and LightGBM by eliminating the need for feature engineering. Unlike its predecessors, TabNet processes tabular data directly without requiring manual feature selection or transformation. This significantly reduces preprocessing efforts and can handle raw, unprocessed data effectively. Details of the model of Tabnet are attached to the supplement file.

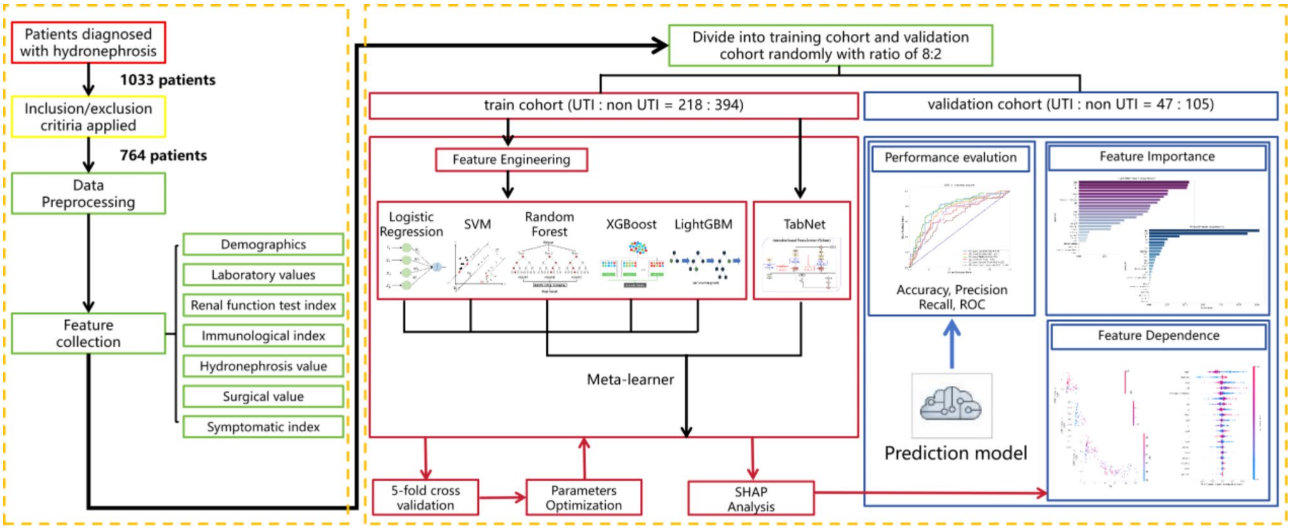


Fig. 2. The technical route of this study.

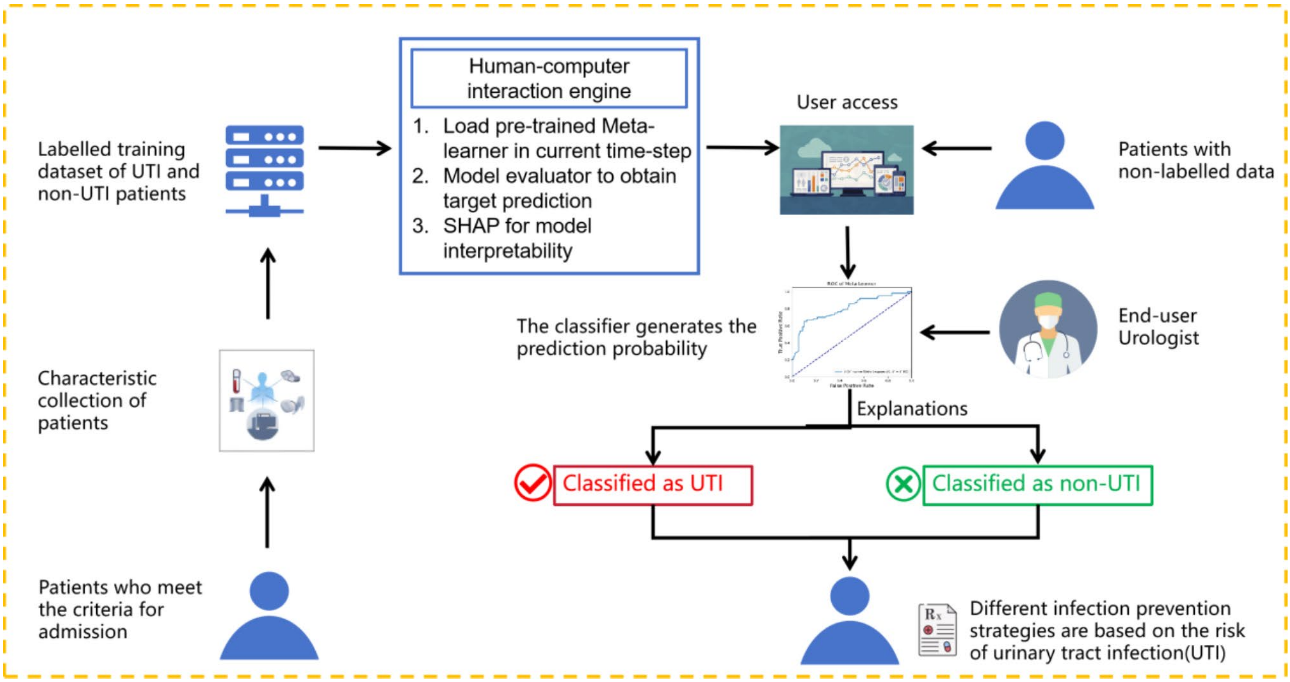


Fig. 3. Application of this study.

Meta learner

However, TabNet's reliance on neural network architecture introduces complexity in model interpretation and requires careful tuning of hyperparameters to achieve optimal performance. Moreover, TabNet is prone to overfitting, especially in small-sample medical scenarios, where the network optimization process may easily get trapped in local minima during training. This can lead to reduced generalization ability and performance degradation on unseen data.

Therefore, in this work, we propose a novel approach named Meta Learner, which introduces an advanced ensemble learning method based on multiple meta models. Meta Learner aims to enhance predictive performance on specific tasks and datasets by dynamically adjusting the contributions of different meta models. Unlike traditional single-model or simple ensemble techniques, Meta Learner integrates cross-model knowledge and achieve knowledge collaboration dynamically.

Specifically, our Meta Learner multi-branch coordinated learning strategy, where each branch comprises independent base models including TabNet, SVM, logistic regression, random forest, LightGBM and XGBoost. Each base model M_i processes input data X independently and generates predictions, as:

$$y_i = M_i(X)$$

These predictions are then combined using a meta-net, which aggregates outputs from all base meta models:

$$\widehat{y} = MetaNet(y1, y2, ?, yn)$$

The meta-model employs a debating ensemble technique, where predictions from each base model are weighted combined to produce a final prediction. This hierarchical learning architecture allows us to capture and utilize information from different abstraction levels, thereby comprehensively understanding data structures and underlying patterns.

SHAP analysis

To further analyze the positive or negative effect of the important features identified for AKI prediction and investigate the relationship between, a shapley additive explanations (SHAP) analysis was performed using Python 3.7.0.

SHAP(Shapley Additive Explanation) is a post-hoc explanation method that draws upon the principles of game theory[32,33]. SHAP evaluates the impact of individual features by calculating their marginal contributions within the model. The fundamental objective of SHAP is to quantify the influence of each feature on the prediction of an instance.

A higher SHAP value indicates a higher probability of AKI occurrence. For each feature, each dot is created according to its positive (red color) or negative (blue color) contribution to the model output of each patient, creating the dots line for each feature of all patients.

Result

General structure of data set

Participants characteristics

The data for this study originated from pediatric patients who underwent surgery for hydronephrosis over a 10-year period at a single center. According to the exclusion criteria, we ultimately included data from 764 patients in a retrospective cohort study (Fig. 1). Of the 764 enrolled cases, 265 experienced urinary tract infections postoperatively, with 217 occurring within 3 months after surgery and 48 occurring between 3 and 6 months postoperatively. Among the 265 urinary tract infection cases, 73 were diagnosed with acute upper urinary tract infections (acute pyelonephritis), and 192 were acute lower urinary tract infections.Among the cases of upper urinary tract infections, 32 were diagnosed with acute lobar bacterial nephritis, 7 with renal abscesses, and new renal scars were detected through DMSA(Dimercaptosuccinic Acid Scintigraphy)scans.

We conducted long-term follow-ups (1 to 8.5 years, median 5.5 years) on each case in this cohort, collecting and recording the occurrence of postoperative urinary tract infections and re-obstructions. The results showed that during the mid-to-long-term follow-up of the entire cohort, 41 cases (5.4%) experienced re-obstruction postoperatively; among these, 22 had previously suffered from urinary tract infections early (within 3 months) after surgery. In contrast, 19 cases (3.8%) in the group without postoperative infections developed long-term re-obstructions, a difference that was statistically significant ($\chi 2 = 155.11$, $P < 0.01$).This demonstrates that postoperative urinary tract infections have a significant positive impact on the recurrence of obstruction at the ureteropelvic junction.

Included features and baseline data

This study enrolled cases that collected 25 clinical characteristics, including demographic characteristics, biochemical renal function test indicators, immunological indicators, and specialized hydronephrosis indicators. The clinical baseline and statistical description of this cohort are shown in Table 1.The training and testing cohorts, which were randomly split in a 4:1 ratio, showed no statistically significant differences in the distribution of the 25 features, as shown in Table 2.

Count data were presented as "cases" and intergroup comparisons were conducted using the *χ2* test or *Fisher's exact probability method*. Measurable data were expressed as *M[Q1*, *Q3]*, and intergroup comparisons were conducted using independent sample t-tests or rank sum tests. *P*-value < 0.05 indicates statistical significance.

Feature engineering and multi-method model

Feature engineering

Traditional machine learning modeling requires feature engineering prior to model establishment. Initially, feature engineering involves ranking the importance of all features based on a baseline model and conducting feature selection. This process includes setting up cross-validation and employing network parameter search methods, typically using accuracy as the criterion. Features that significantly impact the model results are incorporated into the final model. However, deep learning models like TabNet do not necessitate feature selection in feature engineering; they can input all features directly into the final model, only requiring the ranking and analysis of feature importance.

The results of this study indicate that feature engineering can significantly enhance the predictive performance of traditional machine learning models. Taking LightGBM as an example, the test results of the LightGBM model before feature engineering show an accuracy of 0.70, precision of 0.69, recall of 0.70, and an area under the ROC curve of 0.74.After feature engineering, the LightGBM model testing results exhibited an accuracy of 0.71, precision of 0.71, recall of 0.71, and an area under the ROC curve of 0.78.

Variables	Level	Overall	Non-UTI	UTI	*P*	SMD
n		764	499	265		
Gender (%)	F	140 (18.3)	103 (20.6)	37 (14.0)	0.023	0.177
	M	624 (81.7)	396 (79.4)	228 (86.0)		
Surgery (%)	OP	419 (54.8)	310 (62.1)	109 (41.1)	< 0.001	0.43
	LP	345 (45.2)	189 (37.9)	156 (58.9)		
Surgery time (median [IQR])		90.00 [70.00, 120.00]	90.00 [70.00, 120.00]	94.00 [75.00, 120.00]	0.005	0.213
Drainge (%)	Nephrostomy tube	334 (43.7)	279 (55.9)	55 (20.8)	< 0.001	0.776
	D-J tube	430 (56.3)	220 (44.1)	210 (79.2)		
Number of operations (%)	First	26 (3.4)	15 (3.0)	11 (4.2)	0.406	0.062
	Multiple	738 (96.6)	484 (97.0)	254 (95.8)		
Infection (%)	No	722 (94.5)	478 (95.8)	244 (92.1)	0.032	0.156
	Yes	42 (5.5)	21 (4.2)	21 (7.9)		
Fistula (%)	No	740 (96.9)	495 (99.2)	245 (92.5)	< 0.001	0.342
	Yes	24 (3.1)	4 (0.8)	20 (7.5)		
Abdominalgia (%)	No	621 (81.3)	377 (75.6)	244 (92.1)	< 0.001	0.46
	Yes	143 (18.7)	122 (24.4)	21 (7.9)		
Cause (%)	Congenital ureteral stricture	742 (97.1)	484 (97.0)	258 (97.4)	0.391	0.155
	Vagal vein compression	9 (1.2)	7 (1.4)	2 (0.8)		
	Ureteral polyps	9 (1.2)	7 (1.4)	2 (0.8)		
	Ureteral valves	2 (0.3)	1 (0.2)	1 (0.4)		
	High ureter	1 (0.1)	0 (0.0)	1 (0.4)		
	Ureteral rupture	1 (0.1)	0 (0.0)	1 (0.4)		
Side (%)	L	555 (72.6)	374 (74.9)	181 (68.3)	0.05	0.148
	R	209 (27.4)	125 (25.1)	84 (31.7)		
Age (median [IQR])		23.00 [6.39, 60.00]	36.00 [11.00, 72.00]	7.30 [3.90, 24.00]	< 0.001	0.725
Weight (median [IQR])		12.50 [8.50, 20.00]	16.00 [10.00, 23.00]	9.20 [7.50, 13.30]	< 0.001	0.626
Height (median [IQR])		85.00 [68.00, 114.00]	100.00 [74.00, 120.00]	71.00 [65.00, 88.00]	< 0.001	0.723
Bmi (median [IQR])		17.17 [15.51, 19.06]	17.01 [15.28, 18.88]	17.52 [16.00, 19.32]	0.028	0.061
Tp (median [IQR])		65.90 [61.50, 70.73]	67.80 [63.20, 71.90]	62.30 [58.80, 67.20]	< 0.001	0.718
Alb (median [IQR])		43.60 [41.70, 45.60]	43.70 [41.90, 45.50]	43.50 [41.40, 45.90]	0.672	0.03
Glb (median [IQR])		22.40 [17.80, 27.30]	24.20 [20.00, 28.10]	18.20 [16.00, 22.70]	< 0.001	0.84
Ua (median [IQR])		268.00 [224.00, 308.00]	271.00 [230.00, 311.00]	260.00 [216.00, 305.00]	0.021	0.193
Cr (median [IQR])		29.00 [22.60, 37.92]	31.40 [25.00, 39.50]	23.60 [20.40, 30.70]	< 0.001	0.715
Bun (median [IQR])		4.10 [2.80, 5.10]	4.30 [3.20, 5.40]	3.20 [2.29, 4.52]	< 0.001	0.38
eGFR (median [IQR])		36.86 [8.77, 61.30]	49.13 [24.88, 67.63]	10.29 [4.90, 35.57]	< 0.001	0.902
N% (median [IQR])		33.00 [22.08, 45.30]	37.00 [25.60, 46.70]	26.50 [19.00, 41.50]	< 0.001	0.387
L%(median [IQR])		56.95 [44.88, 67.60]	53.00 [42.65, 63.75]	62.80 [48.00, 71.00]	< 0.001	0.375
APD(median [IQR])		3.30 [2.70, 4.00]	3.30 [2.65, 3.80]	3.40 [2.70, 4.30]	0.168	0.131
Renal parenchyma (median [IQR])		2.00 [1.60, 3.00]	2.40 [2.00, 3.40]	1.90 [1.30, 2.50]	< 0.001	0.51

Table 1. Baseline characteristics of the infection group and the non-infection group.

Results of predictive model construction using various algorithms

The performance ranking of the six machine learning and deep learning algorithms indicates that, from lowest to highest, the model performance order is as follows: Logistic Regression, SVM, Random Forest, XGBoost, LightGBM without feature engineering, LightGBM with feature engineering, and finally TableNet, as shown in Table 3; Fig. 4.

	Level	Overall	Training cohort	Validation cohort	*P*	SMD
n		764	612	152		
UTI (%)	No	499 (65.3)	394 (64.4)	105 (69.1)	0.276	0.1
	Yes	265 (34.7)	218 (35.6)	47 (30.9)		
Gender (%)	F	140 (18.3)	118 (19.3)	22 (14.5)	0.17	0.129
	M	624 (81.7)	494 (80.7)	130 (85.5)		
Surgery (%)	OP	419 (54.8)	336 (54.9)	83 (54.6)	0.948	0.006
	LP	345 (45.2)	276 (45.1)	69 (45.4)		
Surgery time (median [IQR])		90.00 [70.00, 120.00]	90.00 [71.00, 120.00]	90.00 [70.00, 120.00]	0.739	0.028
Drainge (%)	Nephrostomy tube	334 (43.7)	272 (44.4)	62 (40.8)	0.416	0.074
	D-J tube	430 (56.3)	340 (55.6)	90 (59.2)		
Number of operations (%)	First	738 (96.6)	589 (96.2)	149 (98.0)	0.277	0.107
	Multiple	26 (3.4)	23 (3.8)	3 (2.0)		
Infection (%)	No	722 (94.5)	579 (94.6)	143 (94.1)	0.798	0.023
	Yes	42 (5.5)	33 (5.4)	9 (5.9)		
Fistula (%)	No	740 (96.9)	590 (96.4)	150 (98.7)	0.149	0.148
	Yes	24 (3.1)	22 (3.6)	2 (1.3)		
Abdominalgia (%)	No	621 (81.3)	497 (81.2)	124 (81.6)	0.917	0.01
	Yes	143 (18.7)	115 (18.8)	28 (18.4)		
Cause (%)	Congenital ureteral stricture	742 (97.1)	593 (96.9)	149 (98.0)	0.217	0.231
	Vagal vein compression	9 (1.2)	9 (1.5)	0 (0.0)		
	Ureteral polyps	9 (1.2)	7 (1.1)	2 (1.3)		
	Ureteral valves	2 (0.3)	2 (0.3)	0 (0.0)		
	High ureter	1 (0.1)	0 (0.0)	1 (0.7)		
	Ureteral rupture	1 (0.1)	1 (0.2)	0 (0.0)		
Side (%)	L	555 (72.6)	447 (73.0)	108 (71.1)	0.623	0.044
	R	209 (27.4)	165 (27.0)	44 (28.9)		
Age (median [IQR])		23.00 [6.39, 60.00]	22.00 [6.30, 60.00]	25.50 [6.60, 71.25]	0.36	0.068
Weight (median [IQR])		12.50 [8.50, 20.00]	12.50 [8.50, 20.12]	13.45 [8.57, 20.00]	0.407	0.059
Height (median [IQR])		85.00 [68.00, 114.00]	85.00 [68.00, 112.25]	86.50 [67.75, 117.25]	0.586	0.05
Bmi (median [IQR])		17.17 [15.51, 19.06]	17.16 [15.51, 19.04]	17.30 [15.55, 19.47]	0.484	0.051
Tp (median [IQR])		65.90 [61.50, 70.73]	65.80 [61.40, 70.50]	66.40 [62.10, 71.73]	0.189	0.115
Alb (median [IQR])		43.60 [41.70, 45.60]	43.55 [41.50, 45.60]	43.65 [42.18, 45.80]	0.16	0.103
Glb (median [IQR])		22.40 [17.80, 27.30]	22.30 [17.60, 27.20]	22.95 [18.30, 27.60]	0.284	0.079
Ua (median [IQR])		268.00 [224.00, 308.00]	268.00 [223.75, 308.00]	270.00 [224.75, 314.75]	0.791	0.046
Cr (median [IQR])		29.00 [22.60, 37.92]	28.75 [22.60, 37.00]	31.00 [23.48, 39.32]	0.102	0.168
Bun (median [IQR])		4.10 [2.80, 5.10]	4.10 [2.76, 5.10]	4.20 [3.08, 5.08]	0.502	0.032
eGFR (median [IQR])		36.86 [8.77, 61.30]	36.02 [8.72, 61.48]	40.80 [9.45, 59.29]	0.607	0.007
N% (median [IQR])		33.00 [22.08, 45.30]	32.20 [22.00, 45.30]	34.35 [23.70, 46.25]	0.462	0.07
L%(median [IQR])		56.95 [44.88, 67.60]	57.00 [45.00, 67.60]	56.85 [42.68, 67.20]	0.722	0.012
Apd (median [IQR])		3.30 [2.70, 4.00]	3.30 [2.60, 4.00]	3.30 [2.80, 4.00]	0.419	0.006
Renal parenchyma (median [IQR])		2.00 [1.60, 3.00]	2.00 [1.60, 3.00]	2.20 [1.78, 3.00]	0.178	0.112

Table 2. Univariate comparison between the training cohort and the validation cohort.

Model	Features	Accuracy	Precision	Recall	AUC of ROC
Logistic regression	18 Features	0.63	0.66	0.63	**0.66**
SVM	18 Features	0.65	0.68	0.65	**0.69**
Random forest	18 Features	0.68	0.68	0.68	**0.72**
XGBoosting	18 Features	0.71	0.71	0.71	**0.75**
LightGBM	25 Features	0.70	0.69	0.70	**0.74**
	18 Features	**0.71**	0.71	**0.71**	**0.78**
TabNet	25 Features	**0.75**	0.86	**0.47**	**0.79**

Table 3. Performance comparison of predictive model construction using different algorithms.

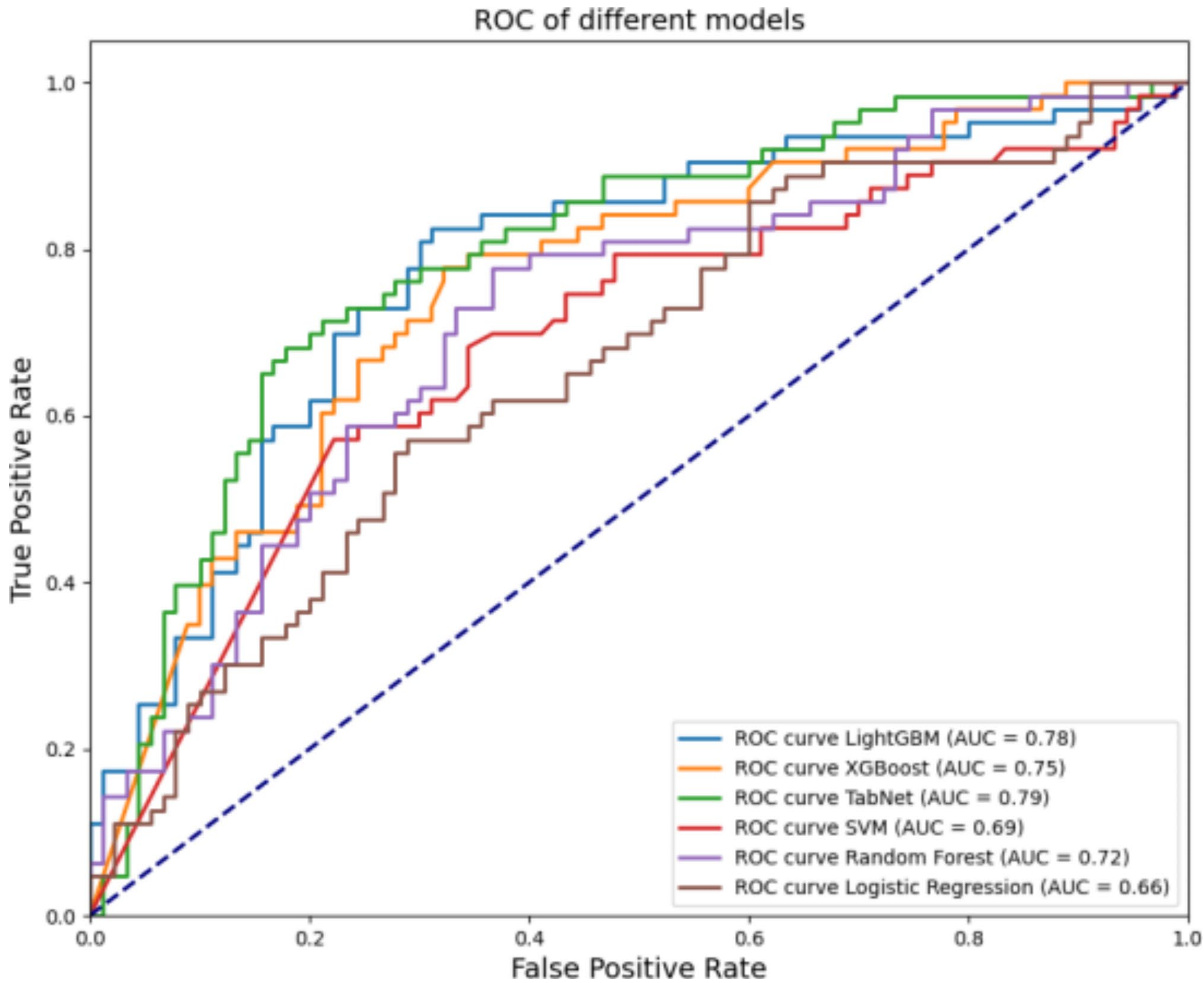
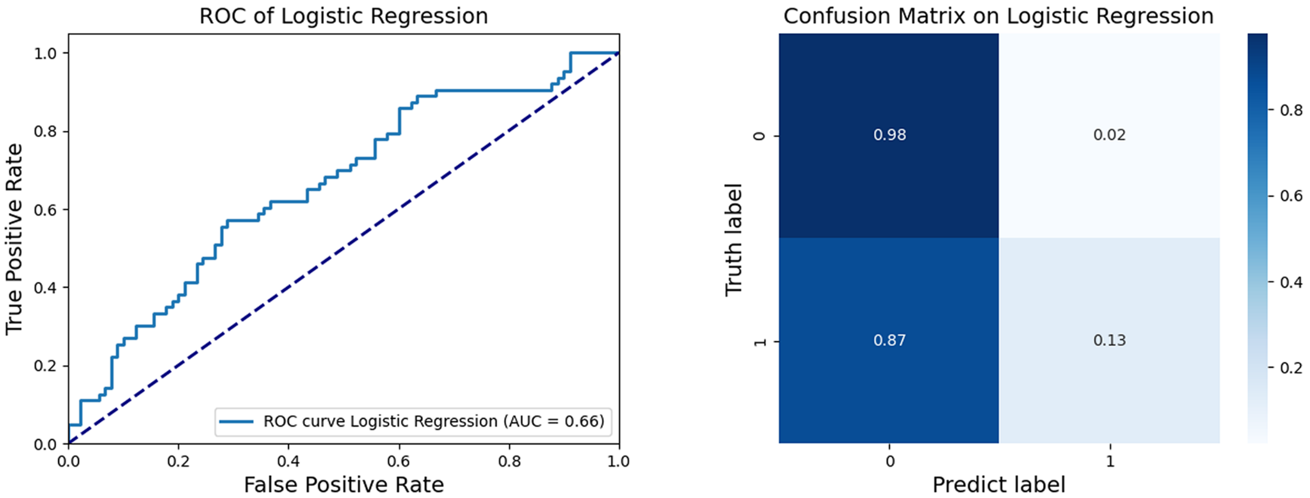
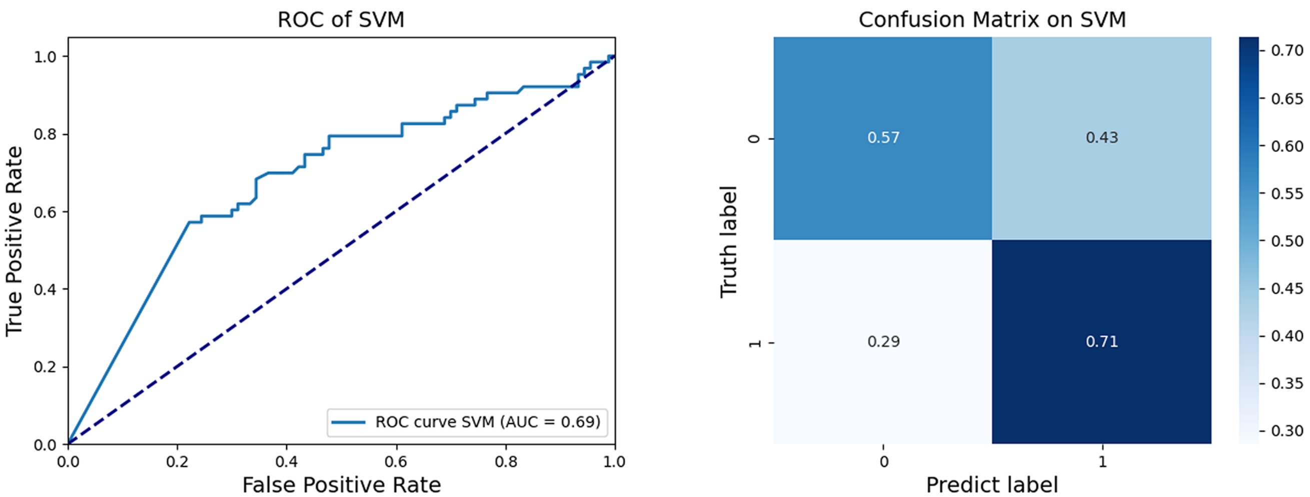


Fig. 4. ROC of different models.

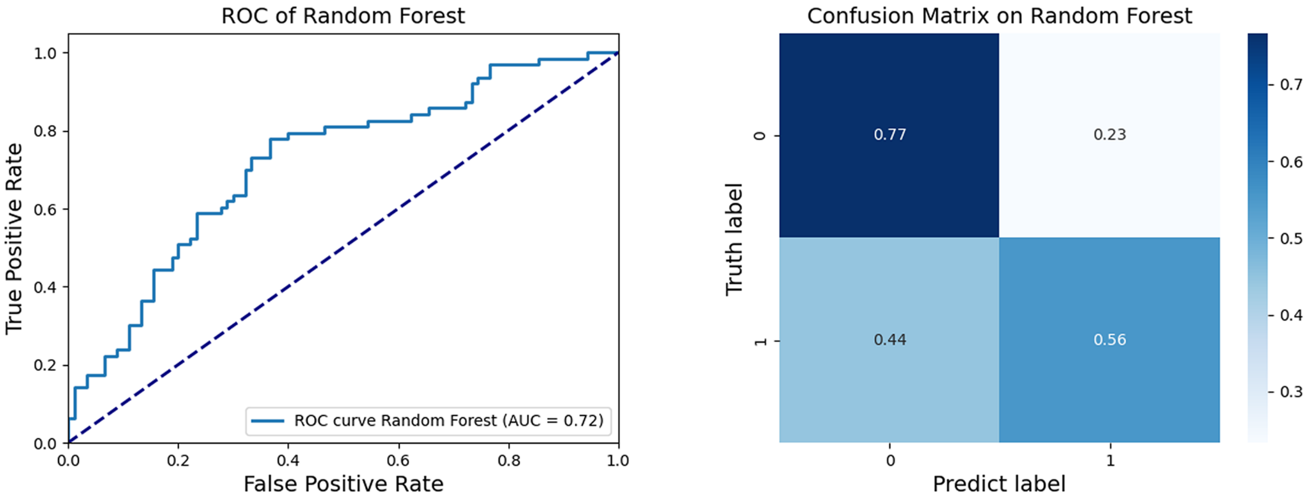
LR:


ROC of Logistic Regression
True Positive Rate
False Positive Rate
ROC curve Logistic Regression (AUC = 0.66)
0.0
0.2
0.4
0.6
0.8
1.0
Confusion Matrix on Logistic Regression
Truth label
Predict label
0
1
0.98
0.02
0.87
0.13
0.2
0.4
0.6
0.8

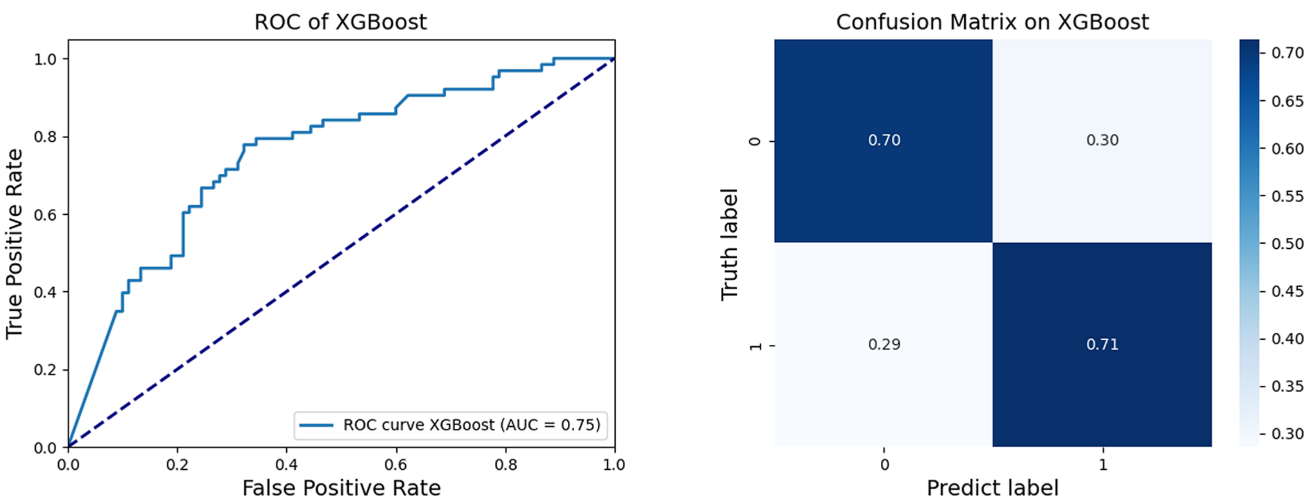
SVM:


ROC of SVM
True Positive Rate
False Positive Rate
ROC curve SVM (AUC = 0.69)
0.0
0.2
0.4
0.6
0.8
1.0
Confusion Matrix on SVM
Truth label
Predict label
0
1
0.57
0.43
0.29
0.71
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70

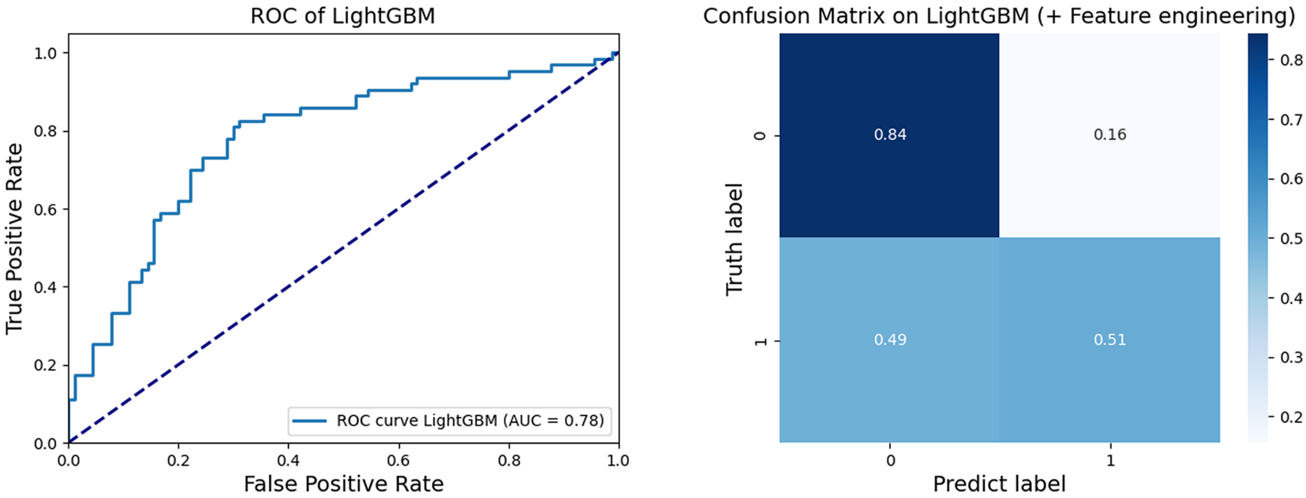
Random forest:



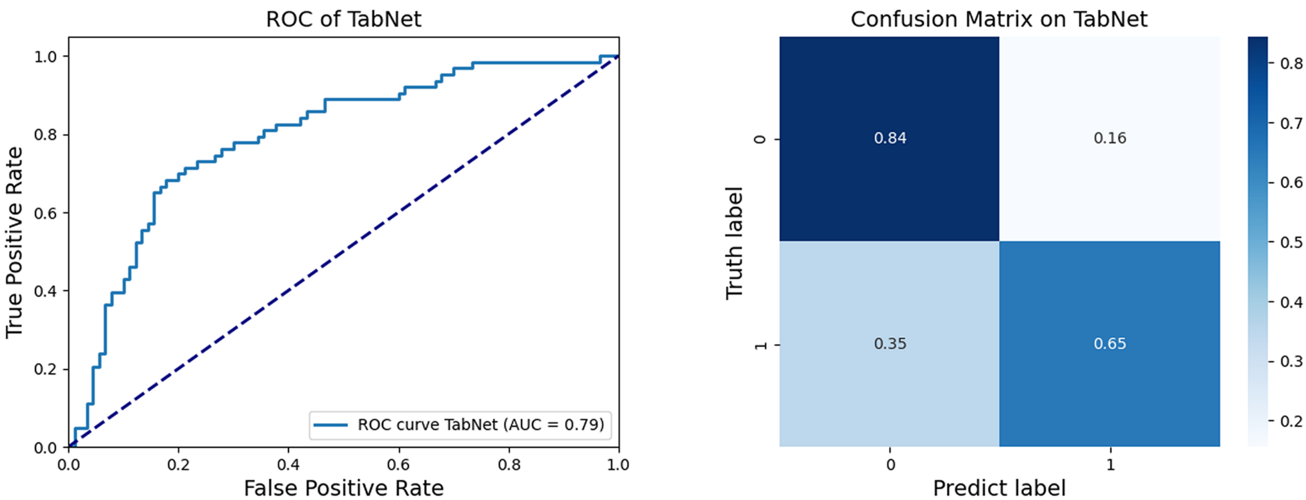
XGBoost:



LightGBM:



TabNet:



The results of modeling with TabNet before and after LightGBM feature engineering

The application of TabNet on the full feature set (25 features) demonstrated superior accuracy, precision, and AUC compared to the LightGBM model; however, it exhibited lower recall. By employing both LightGBM and TabNet as base models and utilizing logistic regression (LR) as a meta-model, we adopted a model ensemble approach to integrate the strengths of these two classifiers. The performance of the ensembled classifier was significantly enhanced compared to the individual base models, with an accuracy of 0.7974, precision of 0.8200, recall of 0.6508, and an ROC AUC score of 0.8018,as shown in Fig. 5.This indicates that the ensemble approach can notably improve the predictive performance, particularly when combining distinct classification algorithms.

The test set performance of LightGBM model (25 features) before feature engineering shows Accuracy: 0.6959, Precision: 0.6933,Recall: 0.6959、ROC AUC: 0.7444.

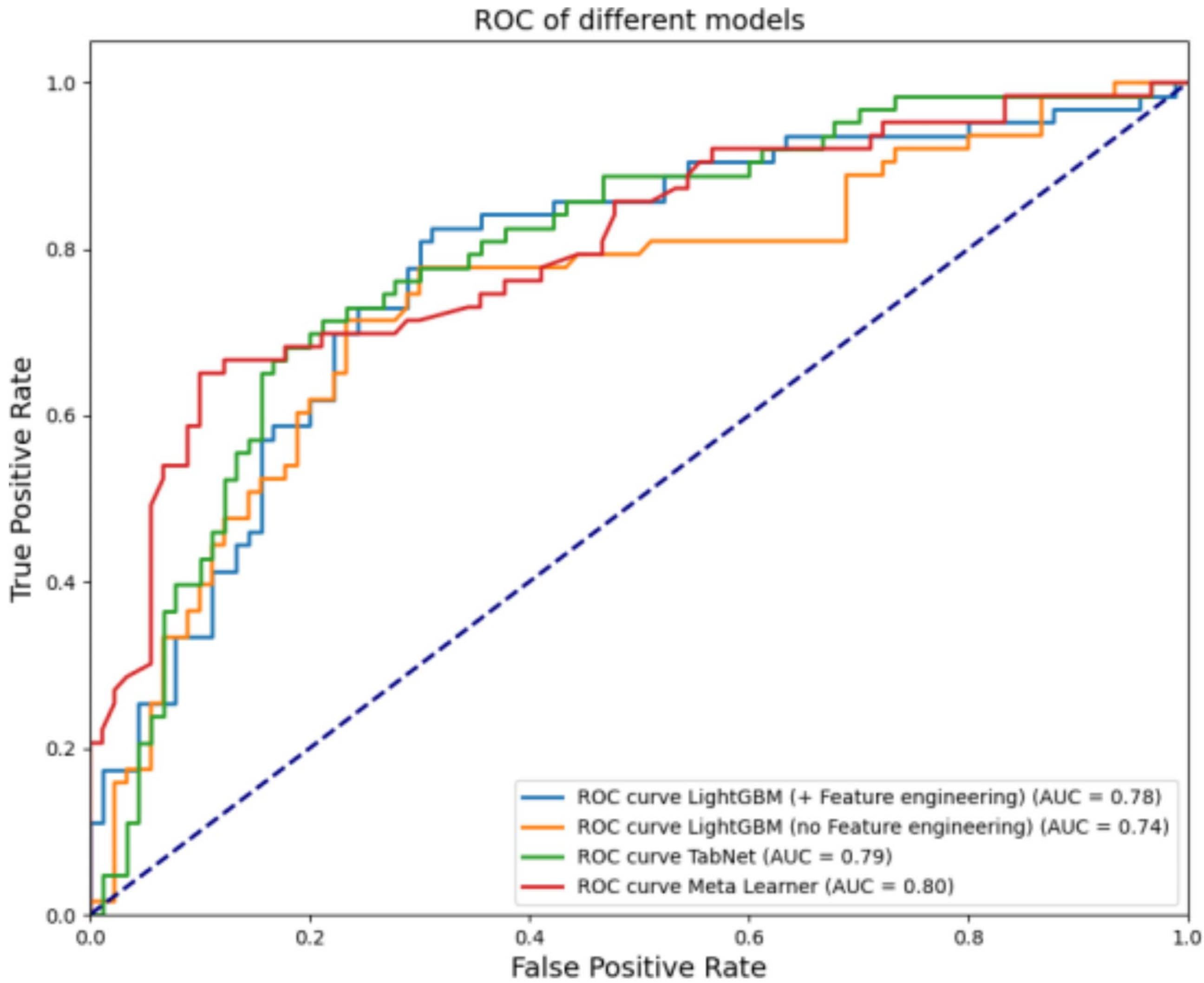
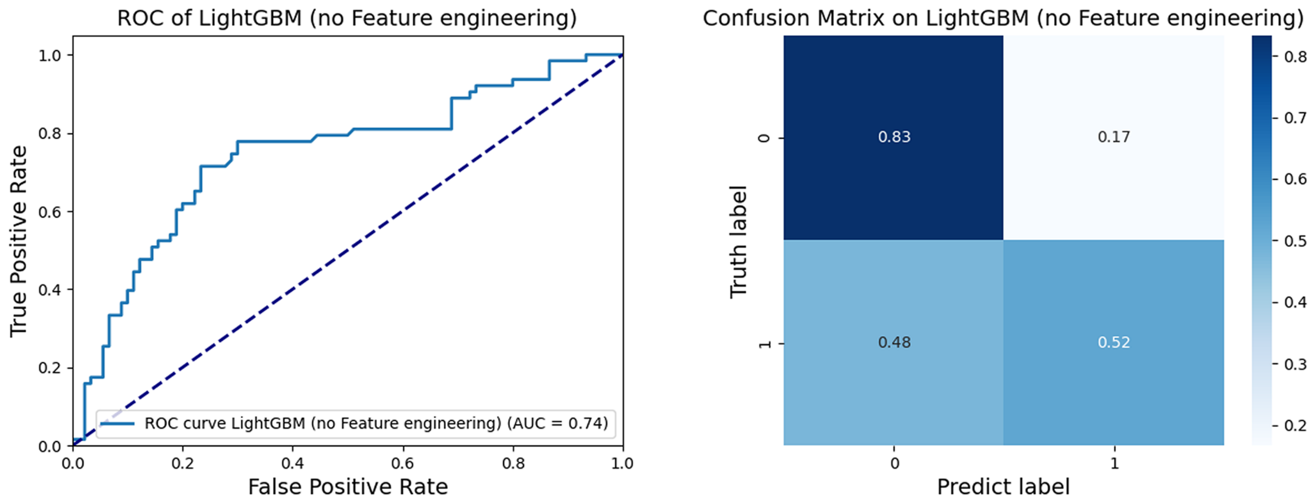
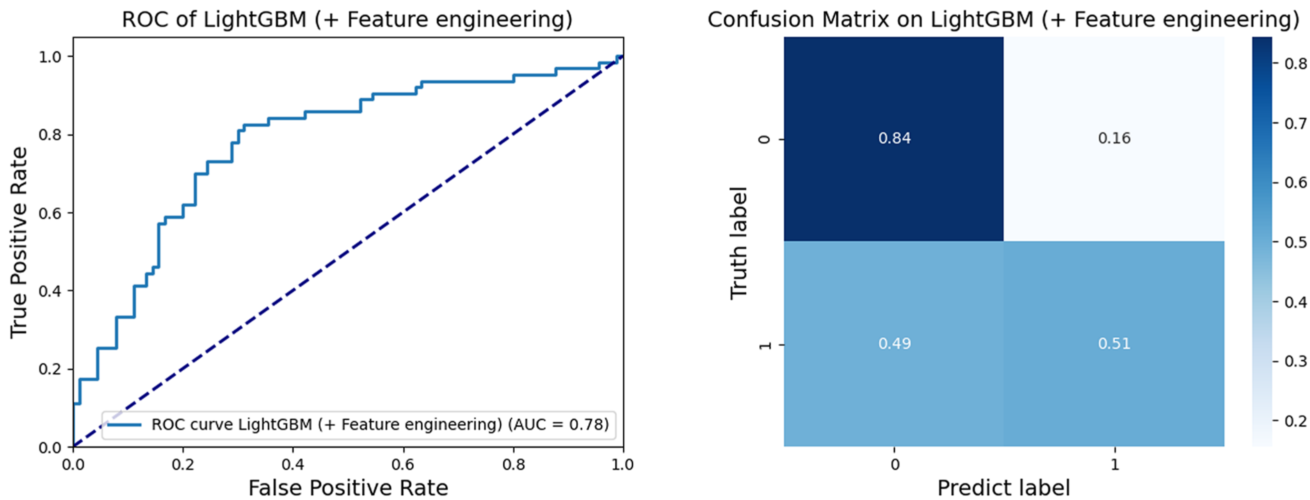


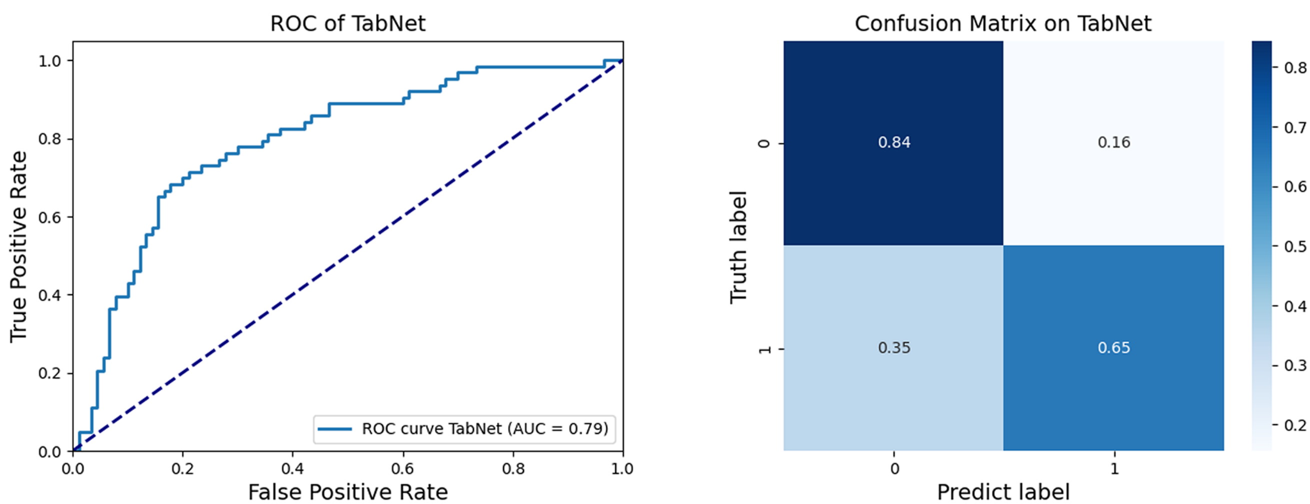
Fig. 5. ROC curves for LightGBM, TabNet, and the Meta-learner.



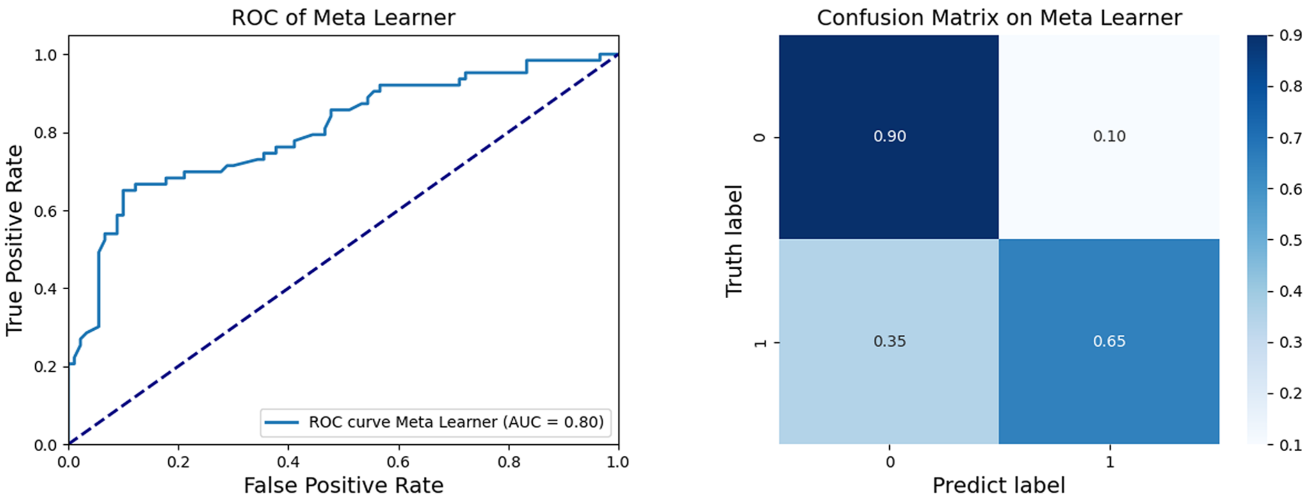
The test set performance of LightGBM model (18 features) after feature engineering shows Accuracy: 0.7059, Precision: 0.7043,Recall: 0.7059、ROC AUC: 0.7794.



The test set performance of Tablenet model (25 features) shows Accuracy: 0.7516, Precision: 0.8571,Recall: 0.4762、ROC AUC: 0.7918.



Lightgbm and tablenet were used as the base model, logistic regression was used as the meta model, and the aggregate model meta learner (25 features) test set performance showed that accuracy: 0.7974 Precision: 0.8200 、Recall: 0.6508、ROC AUC: 0.8018.



Feature Importance Analysis and visualization results of LGB and TabNet

Feature importance analysis of LightGBM

The feature importance results based on the LightGBM model indicate that the estimated Glomerular Filtration Rate (eGFR) is the most impactful feature, followed by Blood Urea Nitrogen (BUN), Serum Albumin (ALB), Neutrophil Ratio (N), Surgical Duration (Time), Serum Creatinine (Cr), Renal Parenchymal Thickness (Renal), Serum Globulin (GLB), Blood Uric Acid (UA), and Lymphocyte Ratio (L). Additionally, Anteroposterior Diameter of the Renal Pelvis (APD), Body Mass Index (BMI), Serum Total Protein (TP), Age (Age), Weight (Weight), Drainage Method (Drainage), Height (Height), and Surgical Approach (Surgery) also have a certain degree of influence in the predictive model,,as shown in Fig. 6.

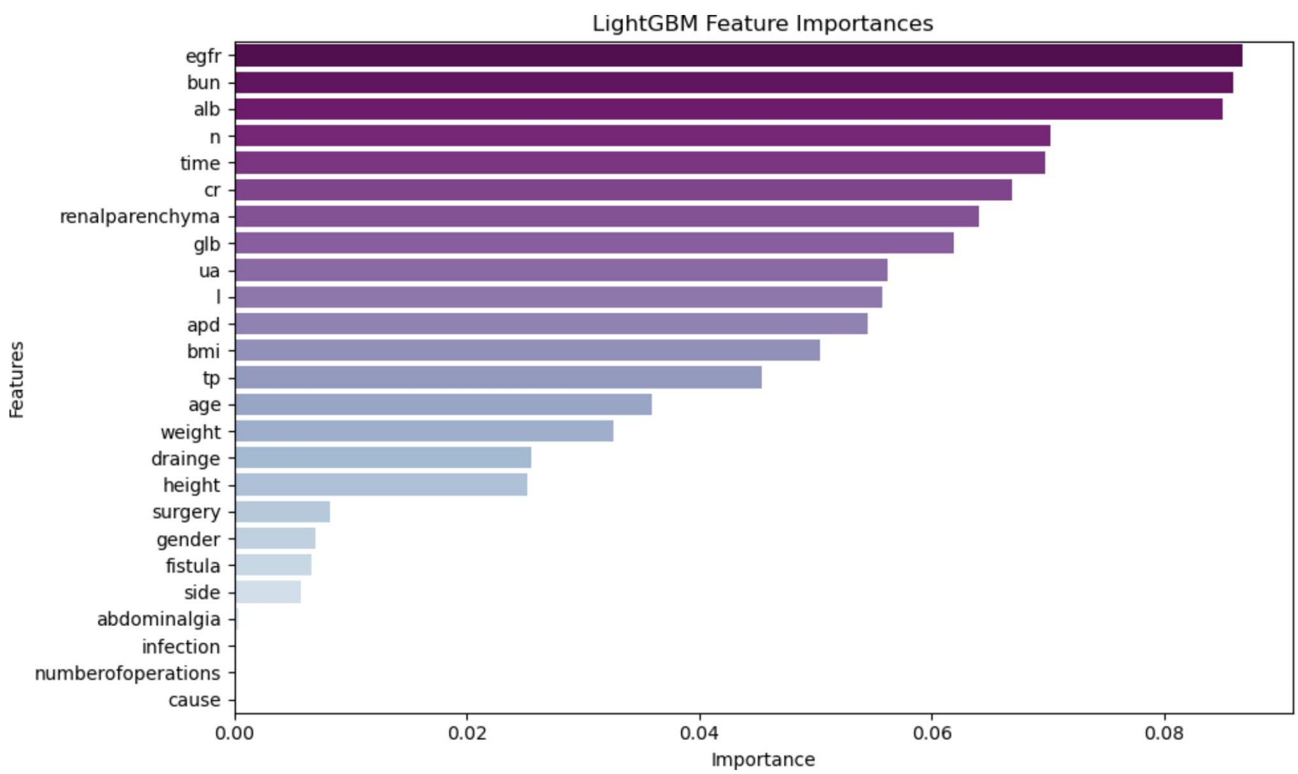


Fig. 6. Feature Importance of LightGBM.

Feature importance analysis of TableNet

The feature importance results based on the TabNet model indicate that Serum Albumin (ALB) is the most impactful feature, followed by Estimated Glomerular Filtration Rate (eGFR), Serum Globulin (GLB), Serum Total Protein (TP), Serum Creatinine (Cr), Lymphocyte Ratio (L), Anteroposterior Diameter of the Renal Pelvis (APD), Drainage Method (Drainage), Surgical Side (Side), and Blood Urea Nitrogen (BUN). Additionally, Age (Age), Body Mass Index (BMI), Height (Height), History of Abdominal Pain (Abdominalgia), Blood Uric Acid (UA), Cause of Pathology (Cause), Number of Operations (NumberofOperations), and Neutrophil Ratio (N) also have a certain degree of influence in the predictive model, as shown in Fig. 7.

Feature importance analysis and feature Intercorrelation after Model Aggregation

Feature importance analysis

The top 20 features based on SHAP values have a significant impact on model output. eGFR is the most influential feature, followed by postoperative drainage method, serum urea nitrogen, serum albumin, renal parenchymal thickness, neutrophil ratio, blood uric acid, serum total protein, surgical duration, serum globulin, serum creatinine, anteroposterior diameter of the renal pelvis, lymphocyte ratio, age, height, BMI index, weight, surgical method, surgical side, and patient gender. To reveal the overall positive or negative impact on model output, we created a SHAP summary plot for the top 20 features of the aggregated model, as shown in the figure. eGFR has a negative correlation with the model prediction results; as eGFR values increase, the trend of decreasing SHAP values indicates that higher eGFR values negatively impact the model predictions. When the postoperative drainage (drainge) is 1 (D-J tube), the SHAP value shows an increasing trend, and as bun values increase, the SHAP value shows a decreasing trend.

Feature dependence analysis

Based on the feature importance scores from the aggregated model, we selected the top four most important features among the top 20 to create a feature dependency plot. The selected features are estimated Glomerular Filtration Rate (eGFR), postoperative drainage method (drainge), serum urea nitrogen (bun), and serum albumin level (alb). As shown in the figures, all three plots demonstrate a trend of decreasing SHAP values as eGFR values increase, indicating that higher eGFR values negatively impact the model predictions. Within the range of low eGFR values, the SHAP values are mostly positive, whereas within the range of high eGFR values, the SHAP values are mostly negative, as shown in Fig. 8.

Figure 9 shows the correlation between different features.**Fig.a** shows the drainage feature, indicating that higher drainage values are more frequent within the range of low eGFR values, whereas lower drainage values are more frequent within the range of high eGFR values. **Fig.b** displays the bun feature and **Fig.c** shows the alb feature, both demonstrating a relatively uniform distribution of bun values and alb values across different ranges of eGFR values. **Fig.d** presents the glb feature, revealing a relatively uniform distribution of glb values across different ranges of drainage method (drainge) values.

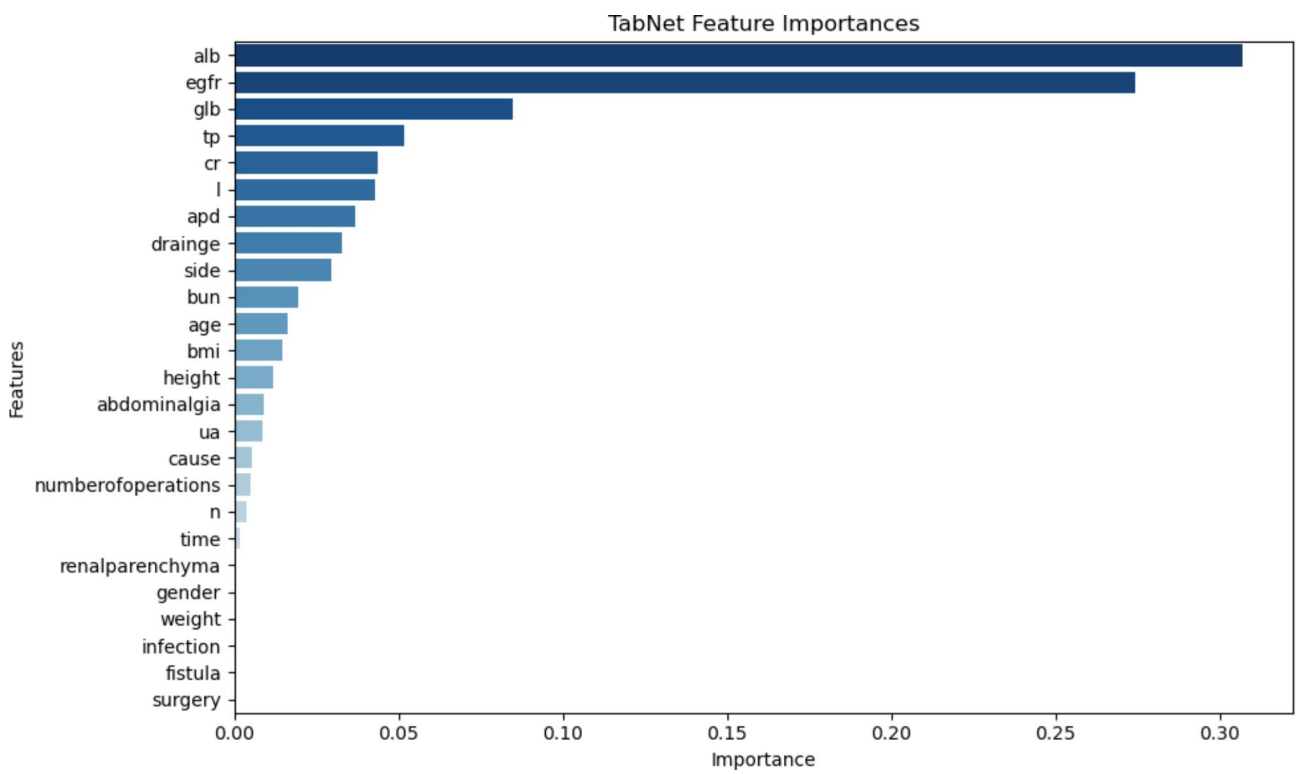


Fig. 7. Feature Importance of TableNet.

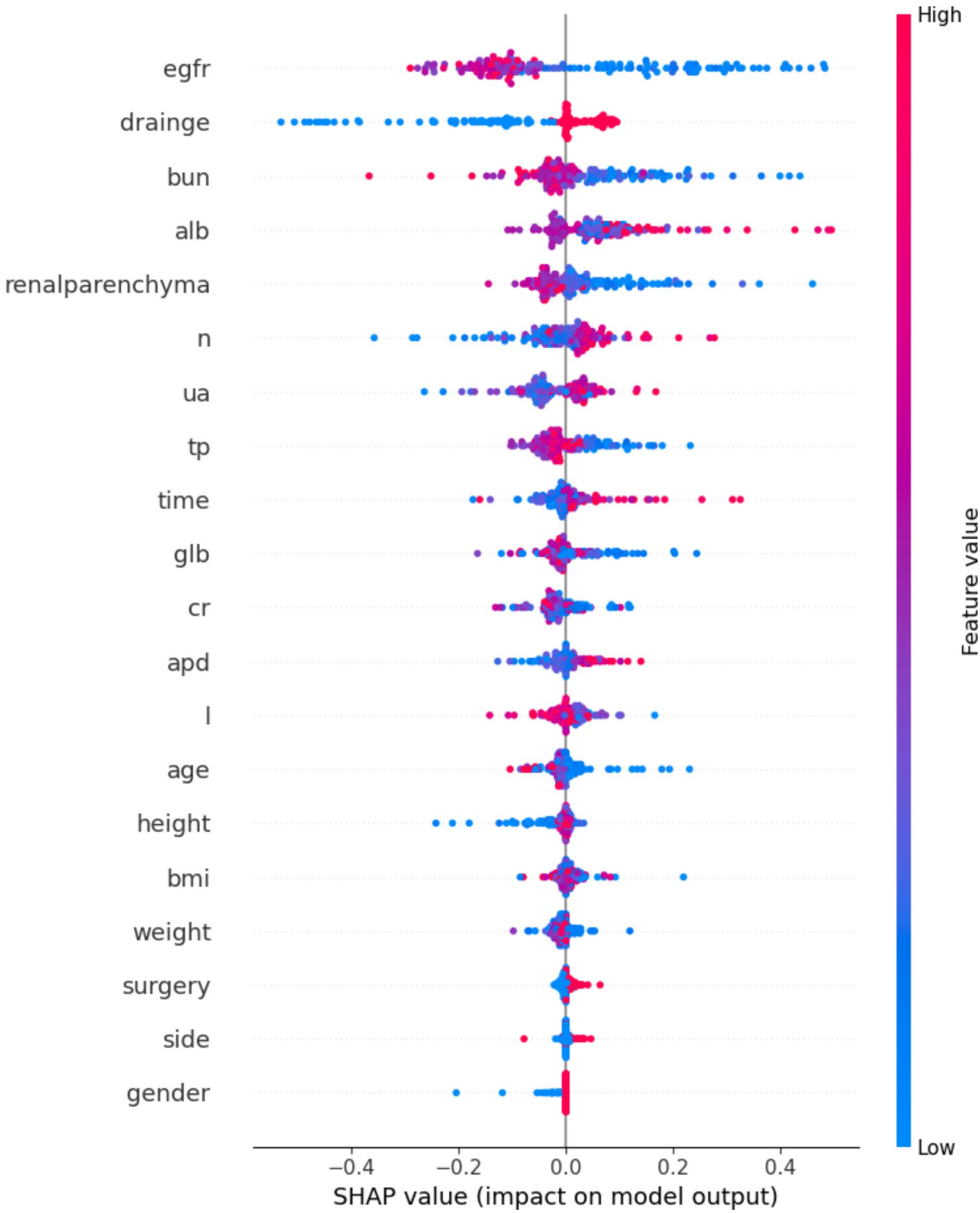


Fig. 8. Feature importance analysis.

Discussion

Traditional machine learning (ML) models, such as Logistic Regression, SVM, Random Forest, XGBoost, and LightGBM, have long been favored for their interpretability and efficiency, particularly in small to medium-sized datasets. These models excel in scenarios where the dataset is well-structured and feature engineering can be systematically applied. The ability to manually craft and select features allows these models to capture relevant patterns in the data, often leading to high predictive performance. However, this reliance on feature engineering is also a significant limitation. The process is not only time-consuming but also requires domain expertise, and

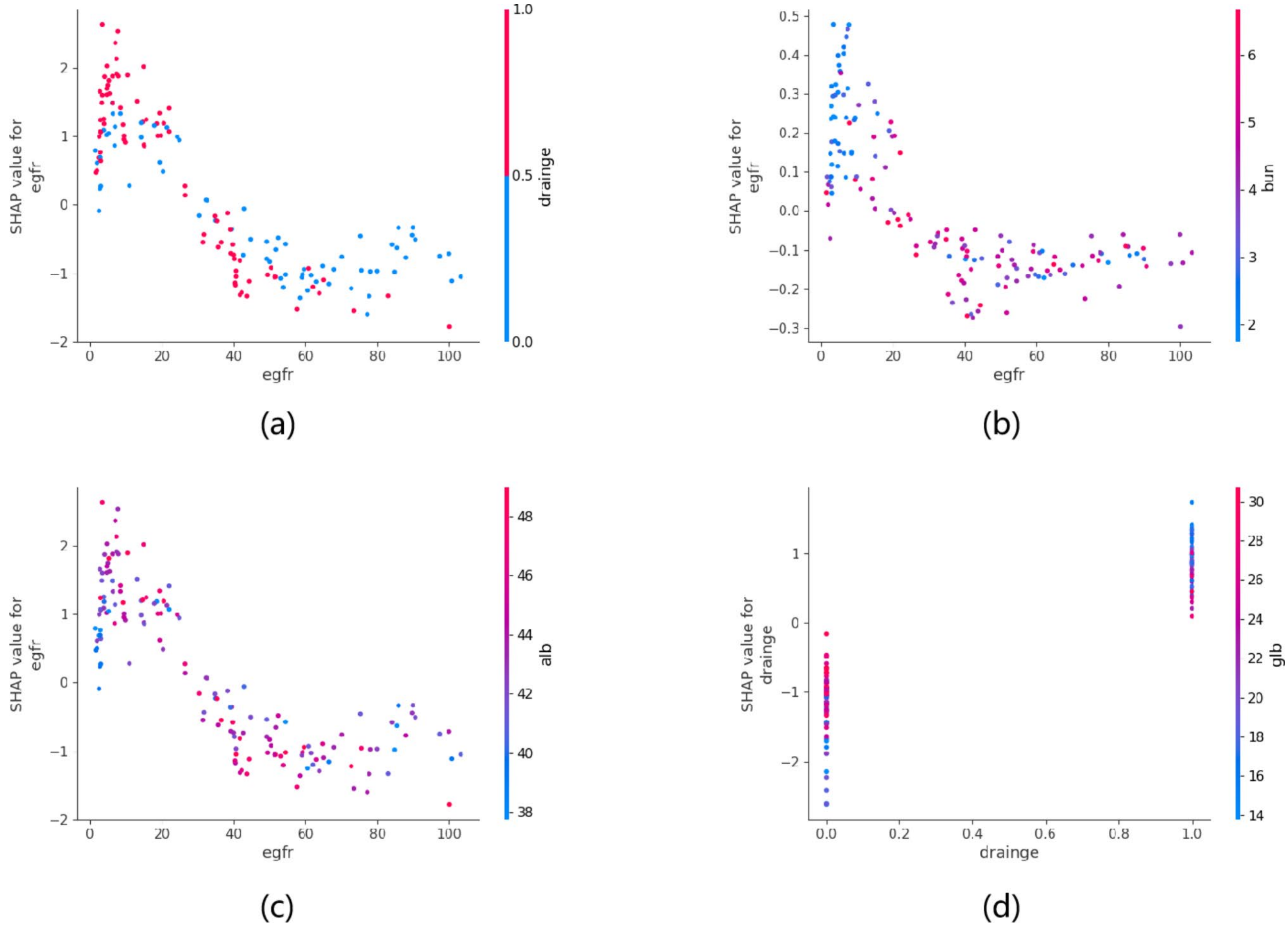


Fig. 9. Feature dependence analysis.

the resulting model performance is highly dependent on the quality and relevance of the engineered features. In complex clinical scenarios, where relationships between variables may be non-linear or subtle, traditional ML models may struggle unless the feature engineering is exceptionally thorough.

Deep neural networks (DNNs), exemplified by models like TabNet, offer a distinct advantage by automating the feature selection process. These models can uncover intricate patterns and interactions within the data that might be missed by traditional ML models, particularly in large datasets. However, in the context of medical research, where datasets are often small due to the rarity of conditions or the difficulty in acquiring data, DNNs face significant challenges. Their tendency to overfit on small samples can lead to biased and unreliable predictions. This is particularly problematic in clinical applications, where generalizability and robustness are critical. The ability of DNNs to perform well without manual feature engineering is offset by their sensitivity to the size and quality of the training data, making them less suitable for small-sample scenarios typical in medical studies.

Recognizing the complementary strengths and weaknesses of traditional ML models and DNNs, we aimed to combine these approaches into a cohesive ensemble learning strategy. We developed the Meta-Learner, an innovative ensemble model that dynamically integrates the predictive capabilities of both ML and DL methods. By leveraging the structured feature engineering strength of traditional ML models alongside the automated, complex pattern recognition of DNNs, our Meta-Learner enhances overall prediction accuracy while mitigating the respective drawbacks of each approach. This model adjusts the contribution of each base model based on the characteristics of the data, effectively balancing the need for robustness in small samples with the ability to uncover complex, non-linear relationships. This hybrid approach not only improves predictive performance but also offers a more generalizable solution for clinical scenarios with small datasets, making it particularly well-suited for medical research applications like predicting UTIs following pediatric pyeloplasty.

To further enhance the interpretability of our model, we conducted a detailed feature importance analysis across different meta-models. The LightGBM model performed feature selection and ranked the importance of the 25 features, which revealed that estimated Glomerular Filtration Rate (eGFR), Blood Urea Nitrogen (BUN), and Serum Albumin Concentration (ALB) were the top three most influential features. Similarly, in the TabNet model, the feature importance analysis identified Serum Albumin Concentration (ALB), estimated Glomerular Filtration Rate (eGFR), and Serum Globulin Concentration (GLB) as the most critical predictors. Notably, both eGFR and ALB consistently ranked as highly important features across both the LightGBM and TabNet models, underscoring their significant role in predicting the risk of postoperative UTIs following pediatric pyeloplasty. This consistency across models highlights the robustness and clinical relevance of these biomarkers in our predictive framework.

The Estimated Renal Glomerular Filtration Rate (eGFR) can be assessed through measuring substance clearance rates, blood circulation markers, and formulas for estimating the glomerular filtration rate. In this study, the eGFR for all cases was calculated using the updated Schwartz bedside eGFR formula (CKiDcr)[34]. Both the LightGBM and TabNet models demonstrated that eGFR had significant importance, contributing highly to the model's predictive power, which holds substantial clinical significance in predicting post-pyeloplasty infections. Previous studies have also confirmed that eGFR can serve as a predictor for urinary tract infections in diabetic patients[35–37]. Although the cases in this study had no history of systemic diseases, considering that all children underwent urological disconnection and reconstruction surgery, which is invasive and increases the risk of urinary tract infections because it damages the urethral mucosa and facilitates bacterial entry, eGFR does not necessarily need to reach a critical value to possess significant clinical relevance[38,39]. Our research results show that the lower the eGFR, the higher the risk of urinary tract infection, and lower levels of eGFR may be associated with the severity of obstructive hydronephrosis. Previous research has indicated that high-risk factors for urinary tract infections include a maximum renal pelvis anterior-posterior diameter > 15 mm, accompanied by dilation of the surrounding renal calyces and changes in the renal parenchyma[40]. Urinary obstruction may also reduce the self-cleaning function of a child's urinary tract, and the kidney system, due to residual excess urine, promotes bacterial growth and reproduction. Studies have shown that urinary retention caused by UPJO provides time and opportunity for bacteria to adhere to urothelium cells, proliferate, and infect the host[41].

In the past, serum albumin levels were commonly regarded as indicators of nutritional status. However, recent literature suggests a close relationship between serum albumin concentration, disease severity, and mortality. Hypoalbuminemia can be considered an independent risk factor for disease incidence and mortality, especially for infectious diseases, where low serum albumin levels are associated with an increased risk of death in patients with severe sepsis[42]. Low albumin concentrations reflect low serum protein levels, leading to reduced bactericidal activity of serum and increased susceptibility to bacterial infections[43]. Research by Falguera et al. considers low serum albumin a predictive factor for infections, such as community-acquired pneumonia[44]. Since the ALB test results in this study were obtained within three days before surgery, the impact of surgery on these values could be minimized. In recent years, some studies have pointed out that albumin levels are also indicators of inflammatory states[45,46]. Research by Lee et al. shows that hypoalbuminemia may reflect baseline nutritional levels or be related to inflammatory changes during the sepsis process[47]. Although the exact mechanism has not been fully elucidated, serum albumin has protective effects, such as antioxidant activity, maintaining physiological homeostasis, and anti-inflammatory actions. Studies by Li et al. have shown that children with acute glomerulonephritis (AGN) who have lower serum protein content are more susceptible to urinary tract infections[48]. In recent years, some scholars have proposed combining different biological test results to predict the risk of urinary tract infections, such as using the procalcitonin/albumin ratio for early diagnosis and prediction of urosepsis and febrile urinary tract infections[49]. In patients with urosepsis, those with a higher procalcitonin/albumin ratio are more likely to develop uroseptic shock[50]. The urea nitrogen/albumin ratio can be used to predict febrile UTI in infants, identify children at high risk of bacteremia, and facilitate management decisions for infants with febrile UTI. The results of this study show that for children undergoing pyeloplasty, attention should be paid to ALB testing levels to early identify postoperative high-risk urinary tract infection cases and take timely clinical management decisions.

While this study has achieved certain results, it still has some limitations. Firstly, the dataset used in this study originates from a single center, and currently lacks support from multi-center validation results. Secondly, the dataset was derived from a retrospective cohort. In the future, we plan to design prospective studies that do not compromise patient welfare, in order to further validate the accuracy of the predictive model and improve prediction precision. The ultimate goal is to reduce infection rates.

Conclusion

This study presents the first ensemble machine learning and deep learning model, specifically designed to predict urinary tract infections (UTIs) following pediatric pyeloplasty. Our findings demonstrate that early postoperative UTIs significantly increase the risk of re-obstruction, emphasizing the importance of accurate prediction and timely intervention. The ensemble approach, which integrates the strengths of machine learning and deep learning model, proved superior in predictive accuracy compared to individual models, achieving high performance without relying heavily on feature engineering. Additionally, SHAP analysis identified eGFR and ALB as key predictive factors, offering new clinical insights into the management of UTI risks in this patient population. This model holds promise for enhancing postoperative care, reducing complications, and improving long-term outcomes in pediatric pyeloplasty. Further research and validation in clinical settings are warranted to confirm its utility and potential for broader application.

Data availability

The desensitized datasets of this study has been provided as a supplementary document.The datasets used and analysed during the current study available from the corresponding author on reasonable request.

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Declarations

Competing interests
The authors have no relevant financial or non-financial interests to disclose.

Ethics approval
Approval was obtained from the ethics committee of the Children's Hospital Capital Institute of Pediatrics (SHERLLM2023041). The procedures used in this study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

Consent to participate
Due to the retrospective nature of this study, we have applied for informed consent in ethical testing.

Additional information
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